



TM254

MANAGING IT: THE WHY, THE WHAT, THE HOW

Module Examination 2026

Tuesday 2nd June 2026

There are **TWO** parts to this paper, and you should attempt **BOTH** parts.

In all parts of the examination paper, the marks for each part of each question are shown: for example (4 marks).

PART 1

Attempt **ALL** questions in Part 1. Part 1 contains eight questions.

Part 1 carries 40 percent of the total examination marks. Each question is worth 5 marks.

PART 2

Attempt **BOTH** questions in Part 2. Part 2 contains two questions.

Part 2 carries 60 percent of the total examination marks. Each question is worth 30 marks.

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PART 1

ANSWER ALL QUESTIONS IN THIS PART.

This part contains eight questions that require short answers requiring no more than a few sentences, or one or two short paragraphs. Each question in Part 1 is worth 5 marks.

Scenario

This scenario is to be used for both of the Part 1 service management questions: question numbers 1 and 2.

Walton Performance is a company that fine-tunes corporate networks to ensure maximum performance from their hardware and systems, and high availability. This is important for organisations such as banks and financial traders where a large amount of streaming data needs to be delivered to dealers, and their decisions communicated quickly. Financial trading is a 24-hour-a-day activity that relies on high-speed communication.

The technical department of Walton Performance is keen to get the most performance out of the equipment for their clients. They have developed a router that is 10% faster than similar products on the market and have been installing these at client sites for the past two years. Recent software improvements have seen the router's performance improve by a further 5%. Still, there is a small risk (approximately 1 in 100,000 at any given time) of the router crashing and becoming unresponsive until it is manually rebooted. Due to the need to manually reboot, the Change Management team has not released the software update universally but has approved it for release to customers who specifically request it.

Question 1

- (a) What value do Walton Performance's customers gain from their relationship with Walton Performance?

(2 marks)

- (b) How is the Change Management team demonstrating good Service Management fundamentals?

(3 marks)

Question 2

Walton Performance decides to create a Premium service tier, which includes the router with the software update, and enhanced support and helpdesk access. As part of the Premium package, it intends to include a Service Level Agreement.

What should be considered when designing the service level agreement?

(5 marks)

Scenario

This scenario is used in questions 3, 4, 5 and 6.

A small pizza chain, Walton Slice, is about to go online. Customers will be able to use an app to select from its menu, supply their delivery details and pay for their order.

Walton Slice has identified the following requirements:

- 1 The website should load within 2 seconds under normal conditions to ensure a smooth user experience.
- 2 Customers should be able to create accounts, log in, view past orders, and save delivery addresses for future use.
- 3 The system must support secure online payments via credit/debit cards and digital wallets.
- 4 The system must allow customers to browse the menu and place orders online, including selecting pizza size, toppings, and sides.
- 5 The system should automatically update ingredient stock levels based on orders placed, alerting staff when stock is low.
- 6 The system should be able to handle increased traffic during peak hours (e.g., Friday evenings) without degradation in performance.
- 7 The interface should be intuitive and mobile-friendly, allowing users to easily navigate and place orders from smartphones.
- 8 Customer data, including payment information, must be encrypted and stored securely in compliance with GDPR.

Question 3

- (a) Identify the functional requirements from the above scenario.
(Just list the numbers of the functional requirements.)

(2 marks)

- (b) Identify the following non-functional requirements from the above scenario
- Scalability
 - Performance
 - Usability

(Just give the type of non-functional requirement and the number from the above scenario.)

(3 marks)

Question 4

Walton Slice wants to ensure that customers' data is secure. It has discussed this with the database developers and has decided it has only three types of employees who will be able to access the database:

- Managers who are trusted with all the application software and direct data access to all the files.
- Order clerks who assist customers with online orders, and therefore need to be able to confirm order details, delivery details, and payment details with customers. They also handle order and delivery preparation.
- And others do things like stock management who do not access the customer data but do need to access summaries of stock levels and stock use, for example.

Briefly describe the features of a database management system (such as Postgres) that will enable the developers to implement a secure database to meet Walton Slice's needs.

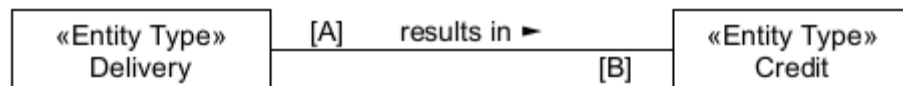
(5 marks)

Question 5

Walton Slice offers has a delivery guarantee. If your order doesn't arrive you get a replacement sent immediately, and you are credited for a deluxe pizza of your choice.

The database keeps track of the failed deliveries and resulting credit. The replacement order is treated by the system as a new order, so if that isn't delivered, it results in its own credit being recorded.

The data analysts have sketched the following diagram:



- (a) Write down two sentences that capture the degree and optionality for the results in the relationship.

(3 marks)

- (b) Complete the following entity relationship diagram, by replacing the [A] and [B] labels with the degree and optionality of the relationship 'results in'.

(2 marks)

Question 6

The following (simplified) tables containing data showing a small number of orders received by Walton Slice, and the pizzas requested on that order.

TABLE: Order

<u>order</u>	customer	delivery address	date	time
352	Fred	6, The Close	2026-4-20	18:45
354	Joan	999, Letsbe Avenue	2026-4-21	21:00
358	Arti	5, The Lane	2026-4-21	20:15

TABLE: OrderLine

<u>order</u>	<u>line</u>	type	size	quantity
352	<u>1</u>	Pepperoni	Medium	2
352	<u>2</u>	Meat Feast	Large	1
354	<u>1</u>	Vegan Feast	Large	1
354	<u>2</u>	Cheese and Ham	Small	3
354	<u>3</u>	Cheesy Does It	Deluxe	1
358	<u>1</u>	Pepperoni	Deluxe	1

For each of the following, **write an SQL query** that answers the request for the Walton Slice database.

- (a) In total, how many Pepperoni pizzas, of any size, have been ordered?

(2 marks)

- (b) For each size of Pepperoni pizzas, how many were needed to satisfy the orders with the Date 2026-4-22? Give the size and number.

(3 marks)

Scenario

This scenario is to be used for Questions 7 and 8.

Walton Veterinary Supplies is a company that specialises in developing portable medical diagnostic toolkits to collect and analyse medical data to diagnose illnesses in horses. These toolkits are used by veterinarians to diagnose illnesses in horses which live in remote areas where travelling to attend a veterinary hospital clinic is difficult, particularly if the horse owners don't have their own horse trailer or horsebox.

Akiko Morita has recently joined Walton Veterinary Supplies as a project manager and will manage a project team developing a gait analysis toolkit to identify the site of lameness in horses. Her team includes not only medical technologists and software developers who work for Walton Veterinary Supplies at their central London laboratories and workshops, but also veterinarians resident elsewhere in the UK who act as both consultants and testers of the toolkits during development.

Akiko has been tasked with developing the toolkit within a year and within a limited budget.

Question 7

As well as money and time, the resources used in a project are often categorised under the following headings:

- Equipment
- Materials
- Space

For each of these headings, identify one resource that is likely to be required for this project, and an associated project risk.

(5 marks)

Question 8

Give two key work priorities for Akiko Morita to consider in order to ensure the project succeeds, explaining why each is important.

(5 marks)

PART 2

Answer BOTH questions from this part.

Each question in Part 2 is worth 30 marks.

Question 9

Walton Research is a campus-based company that rents out physical space and infrastructure support to start-up companies. Its clients rent office space on the campus and have standard office support as part of the rental package. Facilities such as mail delivery, office cleaning, parking, access to a shared canteen and coffee shop are all part of the deal.

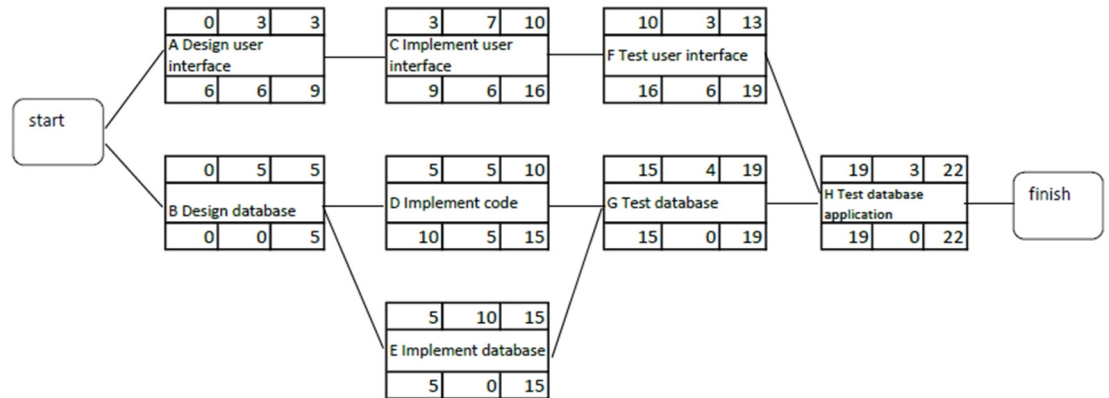
Walton Research also offers a 'digital support' service to its clients. This takes much of the technical infrastructure and support requirements away from the client and centralises the IT systems support. It offers a bring-your-own-device set-up where the clients can attach their own desktops and laptops to the Walton Research network and have full access to serviced local printers, personalised email accounts, website and database server facilities and automated backup and recovery facilities. All of this was fully supported 24 hours a day, 7 days a week by trained technical support personnel. For an additional negotiated per-task fee, Walton Research would also offer an on-demand web design service to support its clients.

Clients reported very positively on the digital support services. It allowed them to focus on their businesses without the overhead of having their own IT department and hardware infrastructure. The clients' customers could contact them on their corporate email addresses, giving a sense of professionalism, and the website design service saved them from hiring contractors or up-skilling their staff to develop and update websites. The printer facilities being serviced centrally meant that there was always someone who could replace the toner cartridges and ensure the paper trays were full. And, of course, automated back-up facilities made it easy to ensure data was not accidentally lost. Facilities such as these were attractive to clients with global web-based businesses.

One year on: Walton Research has been doing well and making changes to services as needed. The campus has been fully occupied for over a year, and the network traffic has grown substantially; so much so that some clients have complained that at peak times, they are noticing a slowdown and poor-quality online meetings. Walton Research decided to suspend the automated backup system when the network becomes busy to reduce the traffic levels. It has also had to reduce the number of staff available 24/7 as costs became too high. So, printers aren't serviced at weekends or between 6 pm and 8 am nightly. There are also times when support staff have been unavailable overnight; on one occasion, this led to a 6-hour outage of all IT services. On a more positive note, they've implemented a more efficient email system that reduced disk storage requirements by over 75%. As part of the implementation, the existing clients had new email addresses successfully issued one week after being informed of the changeover. Demand for website updates and development has also been higher than expected, resulting in a need to book the website design work over two months in advance.

- (a) Identify the services previously enjoyed by Walton Research's clients, and indicate the value provided for each to the user.
(4 marks)
- (b) Give examples of how the clients' risk levels will have increased because of some of the changes made by Walton Research.
(4 marks)
- (c) Identify the loss of value to the customers, one year on, as a result of the changes made by Walton Research.
(4 marks)
- (d) Which of the changes made, one year on, would suggest that there was poor consideration of the following during service design? Explain your answer.
- i. Capacity management
 - ii. Service continuity management
 - iii. Availability management
- (9 marks)
- (e) What is a Business Impact Analysis, and how would one have helped Walton Research better understand the effect of the changes on its clients?
(5 marks)
- (f) The email change is seen by the scenario writer as positive news – a disk storage reduction of over 75%, and all email addresses being changed within a week. How does this suggest the writer is not aware of service management aims?
(4 marks)

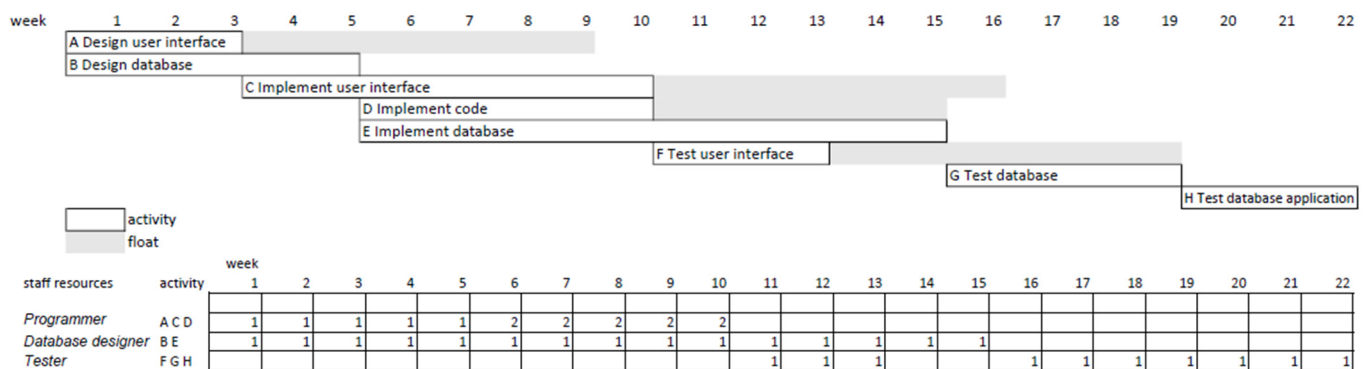
Question 10 Below is a precedence network for developing a database application where the time periods are in weeks.



The table below shows the number and type of staff required for each activity.

Activity	Resource	Number of weeks	Quantity
A Design user interface	Programmer	3	1
B Design database	Database designer	5	1
C Implement user interface	Programmer	7	1
D Implement code	Programmer	5	1
E Implement database	Database designer	10	1
F Test user interface	Tester	3	1
G Test database	Tester	4	1
H Test database application	Tester	3	1

Below is the activity bar and chart resource table for the project.



(a) What is the peak resource, and which week(s) does it occur?

(2 marks)

(b) Suppose time is a constraint in this project, and there is no flexibility in extending the project deadline. Identify and explain *one* possible strategy for resource smoothing.

(8 marks)

- (c) For each of the scenarios below, suggest possible solutions and briefly discuss the factors that need to be considered when deciding the course of action to take, including staff allocations and project cost.
- i. After completing activity B *Design database*, the database designer who was allocated to that activity was diagnosed with an acute viral infection and will need to be off work for between 1 to 2 weeks to recover.
 - ii. At the start of the project, it was found that the estimate of 3 weeks for activity A, *Design user interface* and 7 weeks for activity C, *Implement user interface*, was far too optimistic, and these activities are now estimated to take 6 and 10 weeks, respectively.

(20 marks)

[END OF QUESTION PAPER]